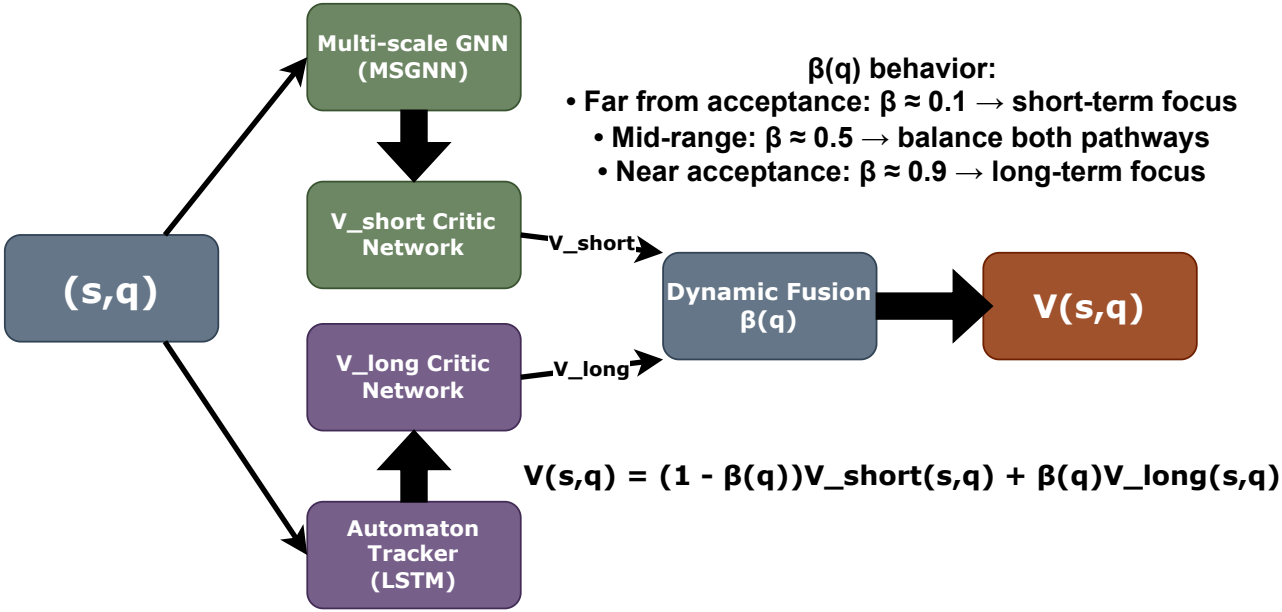
Multi-scale GNN
(MSGNN)
(s,q)
V_short Critic
Network
V_long Critic
Network
Automaton
Tracker
(LSTM)
V_short
V_long
Dynamic Fusion
β(q)
V(s,q)
β(q) behavior:
• Far from acceptance: β ≈ 0.1 → short-term focus
• Mid-range: β ≈ 0.5 → balance both pathways
• Near acceptance: β ≈ 0.9 → long-term focus
V(s,q) = (1 - β(q))V_short(s,q) + β(q)V_long(s,q)